

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Title – JanSahayak AI: A Multi Agent AI-powered Government Scheme Recommendation and Citizen Assistance System

Domain of Project – E-Governance

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Problem Statement - JanSahayak AI

The Challenge:

Millions of citizens are unaware of the government welfare schemes for which they are eligible. Existing government portals often require users to manually search through multiple websites, understand complex eligibility criteria, collect numerous documents, and complete lengthy application procedures. This leads to low awareness, missed opportunities, delayed access to benefits, and underutilization of welfare programs, particularly among rural populations, students, farmers, women, senior citizens, and economically weaker sections.

Traditional systems provide static information but lack **personalized recommendations, intelligent eligibility assessment, multilingual support, proactive reminders, and application guidance**. Citizens need an AI-powered assistant that simplifies access to government services through personalized, real-time, and evidence-based recommendations.

The Objective:

Build an **AI-powered Multi-Agent Government Scheme Recommendation and Citizen Assistance System** that delivers **personalized government scheme recommendations and end-to-end citizen support** using IBM watsonx.ai (IBM BOB) and IBM Cloud.

The solution should:

- **Personalized Citizen Profiling** – Collect and analyze citizen information such as age, gender, education, occupation, income, location, social category, disability status, and family details to identify relevant government schemes.
- **Government Scheme Retrieval** – Use **Retrieval-Augmented Generation (RAG)** to retrieve accurate and up-to-date information from trusted Central and State Government sources.
- **Eligibility Verification** – Automatically evaluate eligibility criteria for multiple schemes and identify the most suitable options.
- **Scheme Recommendation Agent** – Rank and recommend schemes based on citizen needs, benefits, and eligibility.

Proposed Solution - **JanSahayak AI**

The proposed solution is a **multi-agent AI-powered Government Scheme Recommendation System** built using **IBM BOB, IBM Watson, IBM Granite Model and IBM Cloud** to provide personalized, intelligent, and real-time government welfare scheme recommendations. The system addresses challenges in citizen awareness, eligibility verification, application guidance, and access to government services through an intelligent, modular Agentic AI architecture.

The system consists of the following **Four AI agents**:

Citizen Profile Agent Collects and analyses citizen information such as age, gender, education, occupation, annual income, state, district, social category, disability status, and family details to create a comprehensive user profile.

Government Scheme Retrieval Agent uses to retrieve relevant Central and State Government schemes from trusted knowledge sources and summarize their key features in simple language.

Eligibility Verification Agent evaluates citizen information against scheme-specific eligibility criteria and identifies all applicable welfare schemes automatically.

Scheme Recommendation Agent ranks and recommends the most suitable schemes based on the citizen's profile, priorities, financial needs, and expected benefits.

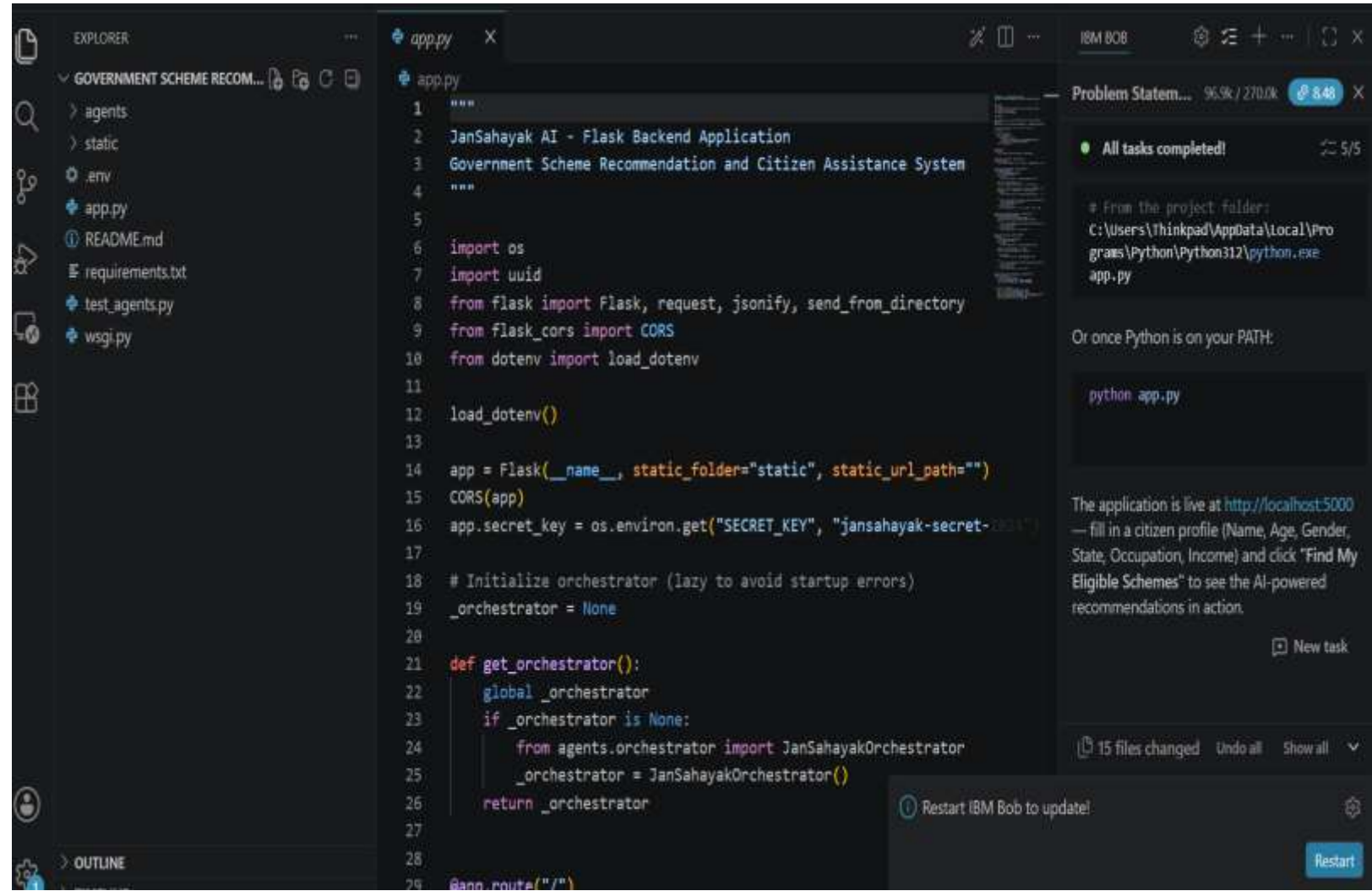
Overall, this solution delivers an **end-to-end Agentic AI-powered citizen assistance platform** that simplifies access to government welfare schemes, improves citizen awareness, automates eligibility verification, streamlines the application process, and promotes inclusive, transparent, and efficient digital governance through personalized recommendations and intelligent decision support.

Technology Used- **JanSahayak AI**

- **IBM BOB Platform:** For designing and orchestrating multi agent workflows visually.
- **IBM Granite model:** IBM-Granite-4-0-h-small instruct-For natural language understanding, reasoning, and personalization
- **IBM Cloud:** It provides the infrastructure to build, deploy, and scale the JanSahayak AI efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the JanSahayak AI Agent.
- **IBM Watson studio, Watson Runtime, Cloud object Storage:** Creating a new project on IBM Cloud.
- **Prompt Engineering:** For defining the goal of the Agent.

Required Files created using IBM bob – JanSahayak AI

1. App.py
2. Requirements.txt
3. .env
4. test_agents.py
5. wsgi.py



IBM Bob for JanSahayak AI

The screenshot displays the IBM Bob IDE interface for a project titled "Government Scheme Recommendation". The interface is divided into several panels:

- EXPLORER (Left):** Shows the project structure with folders "agents" and "static", and files ".env", "app.py", "README.md", "requirements.txt", "test_agents.py", and "wsgi.py".
- Central Editor:** Displays a large, stylized logo of a robot wearing a blue hard hat and a white shirt with blue code symbols "</>". Below the logo, there are keyboard shortcuts: "Show All Commands" (Ctrl + Shift + P), "Go to File" (Ctrl + P), and "Open Settings" (Ctrl + ,).
- Top Bar:** Includes a menu bar (File, Edit, Selection, ...), navigation arrows, the project name "Government Scheme Recommendation", and buttons for "Update" and "Upgrade your plan".
- Right Panel (Task View):**
 - Shows the "IBM BOB" logo and a "Problem Statement..." section with a progress bar (96.9k / 270.0k) and a score of 8.48.
 - A green status indicator says "All tasks completed!" with a progress bar (5/5).
 - Instructions: "# from the project folder: C:\Users\Thinkpad\AppData\Local\Programs\Python\Python312\python.exe app.py".
 - Text: "Or once Python is on your PATH: python app.py".
 - Information: "The application is live at [http://localhost:5000](\"http://localhost:5000\") — fill in a citizen profile (Name, Age, Gender, State, Occupation, Income) and click 'Find My Eligible Schemes' to see the AI-powered recommendations in action."
 - A "New task" button.
 - A notification: "15 files changed" with "Undo all" and "Show all" options.
 - A "Restart IBM Bob to update!" notification with a "Restart" button.
- Bottom Bar:** Shows "0 0 0" (likely error/warning counts) and "Bob Findings".

IBM Bob for JanSahayak AI

The image shows a VS Code editor with a project named "GOVERNMENT SCHEME RECOM..." and a file explorer on the left. The main editor displays the code for `app.py`, which is a Flask backend application. The code includes imports for `os`, `uuid`, `Flask`, `request`, `jsonify`, `send_from_directory`, `flask_cors`, `CORS`, and `dotenv`. It initializes a Flask app with `static_folder="static"` and `static_url_path=""`, sets a secret key, and initializes an orchestrator. A `get_orchestrator()` function is defined to return the orchestrator. The code ends with a `@app.route("/")` decorator.

On the right, the IBM Bob interface is visible. It shows a "Problem Statement..." tab with a progress bar at 96.9% / 270.0k and a score of 8.46. Below this, it states "All tasks completed!" and provides instructions on how to run the application. It suggests running `python app.py` from the project folder or once Python is on the PATH. It also provides the URL `http://localhost:5000` and instructions on how to use the application. At the bottom, it says "15 files changed" and "Undo all" and "Show all".

```
1 """
2 JanSahayak AI - Flask Backend Application
3 Government Scheme Recommendation and Citizen Assistance System
4 """
5
6 import os
7 import uuid
8 from flask import Flask, request, jsonify, send_from_directory
9 from flask_cors import CORS
10 from dotenv import load_dotenv
11
12 load_dotenv()
13
14 app = Flask(__name__, static_folder="static", static_url_path="")
15 CORS(app)
16 app.secret_key = os.environ.get("SECRET_KEY", "jansahayak-secret-3624")
17
18 # Initialize orchestrator (lazy to avoid startup errors)
19 _orchestrator = None
20
21 def get_orchestrator():
22     global _orchestrator
23     if _orchestrator is None:
24         from agents.orchestrator import JanSahayakOrchestrator
25         _orchestrator = JanSahayakOrchestrator()
26     return _orchestrator
27
28
29 @app.route("/")
```

IBM BOB

Problem Statement... 96.9% / 270.0k 8.46

All tasks completed! 5/5

From the project folder:
C:\Users\Thinkpad\AppData\Local\Programs\Python\Python312\python.exe
app.py

Or once Python is on your PATH:

python app.py

The application is live at <http://localhost:5000>
— fill in a citizen profile (Name, Age, Gender, State, Occupation, Income) and click "Find My Eligible Schemes" to see the AI-powered recommendations in action.

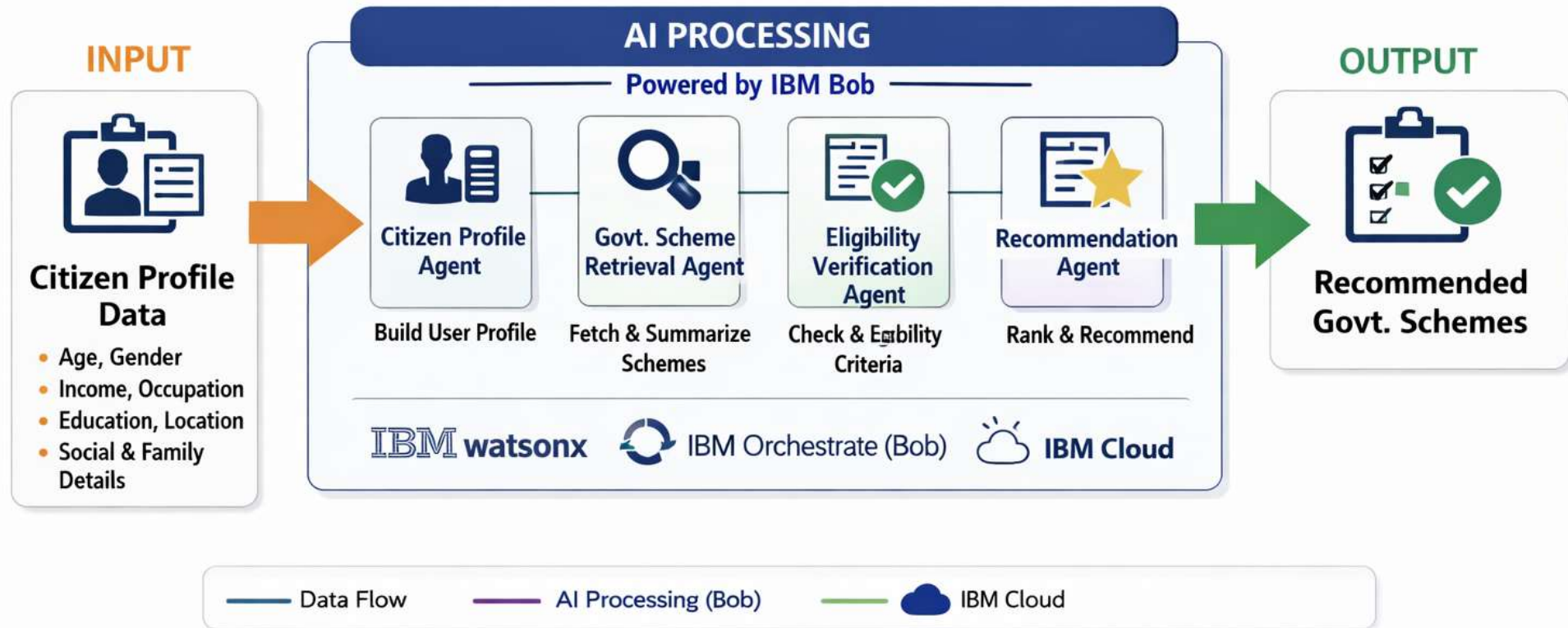
New task

15 files changed Undo all Show all

Restart IBM Bob to update!

Restart

Architecture Blueprint for JanSahayak AI for Government Scheme Recommendation System using Bob



Role of Agentic AI in the solution

Role of Agentic AI in JanSahayak AI Agent

Agentic AI enables **JanSahayak AI** to function as an intelligent, autonomous citizen assistant rather than a simple chatbot. Using **IBM BOB, IBM Watsonstudio and IBM Granite Models**, it orchestrates multiple specialized AI agents to deliver personalized government scheme recommendations and application support.

Each agent performs a dedicated task—such as **citizen profiling, government scheme retrieval, eligibility verification, recommendation generation, document assistance, and application guidance**—while collaborating to provide accurate and reliable services. This **multi-agent architecture** improves efficiency, decision-making, and user experience.

Agentic AI provides **context awareness** by understanding citizen details such as age, income, occupation, education, location, social category, and family information to recommend the most relevant welfare schemes. It also enables **real-time adaptation**, continuously updating recommendations as user profiles, government policies, and new schemes change.

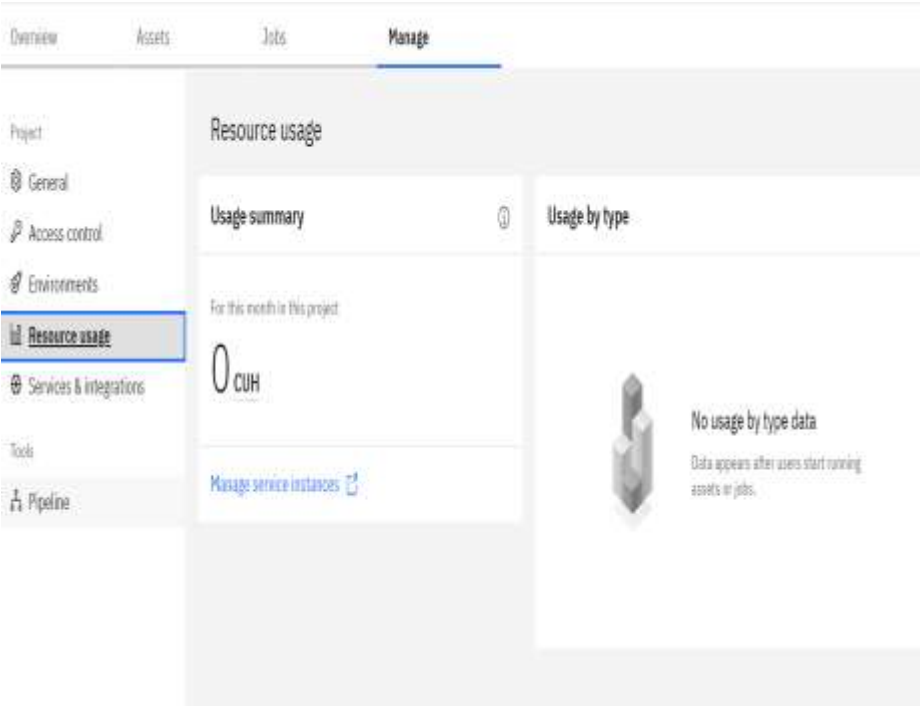
Overall, Agentic AI transforms **JanSahayak AI** into a smart digital citizen assistant that proactively recommends government schemes, verifies eligibility, simplifies applications, and improves access to public welfare services.



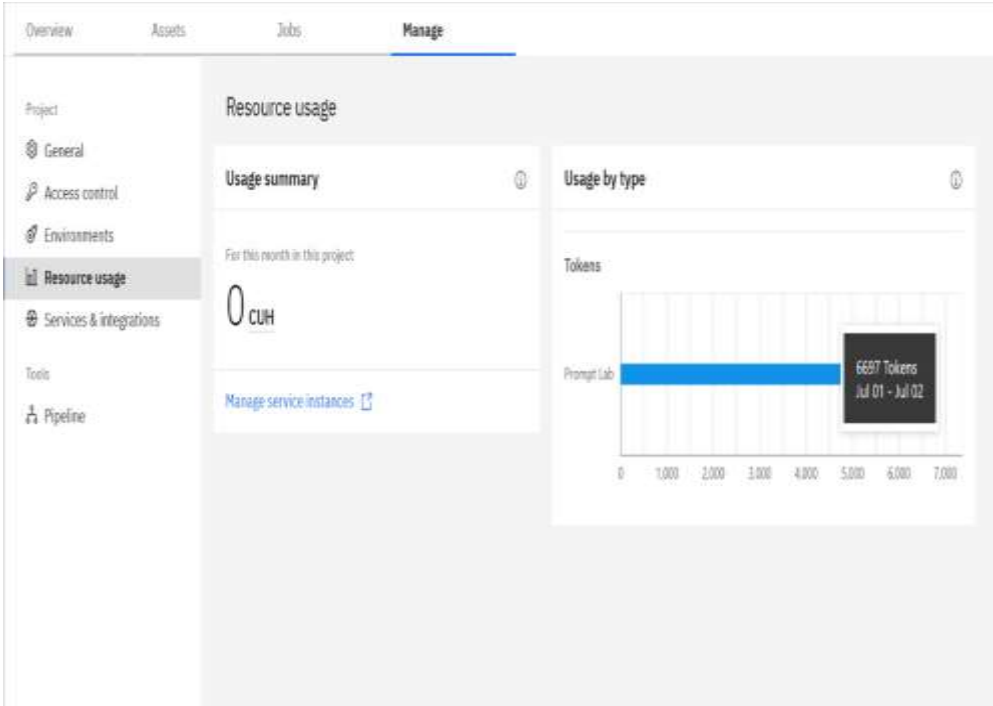
Token Usage- JanSahayak AI

6697 token are used

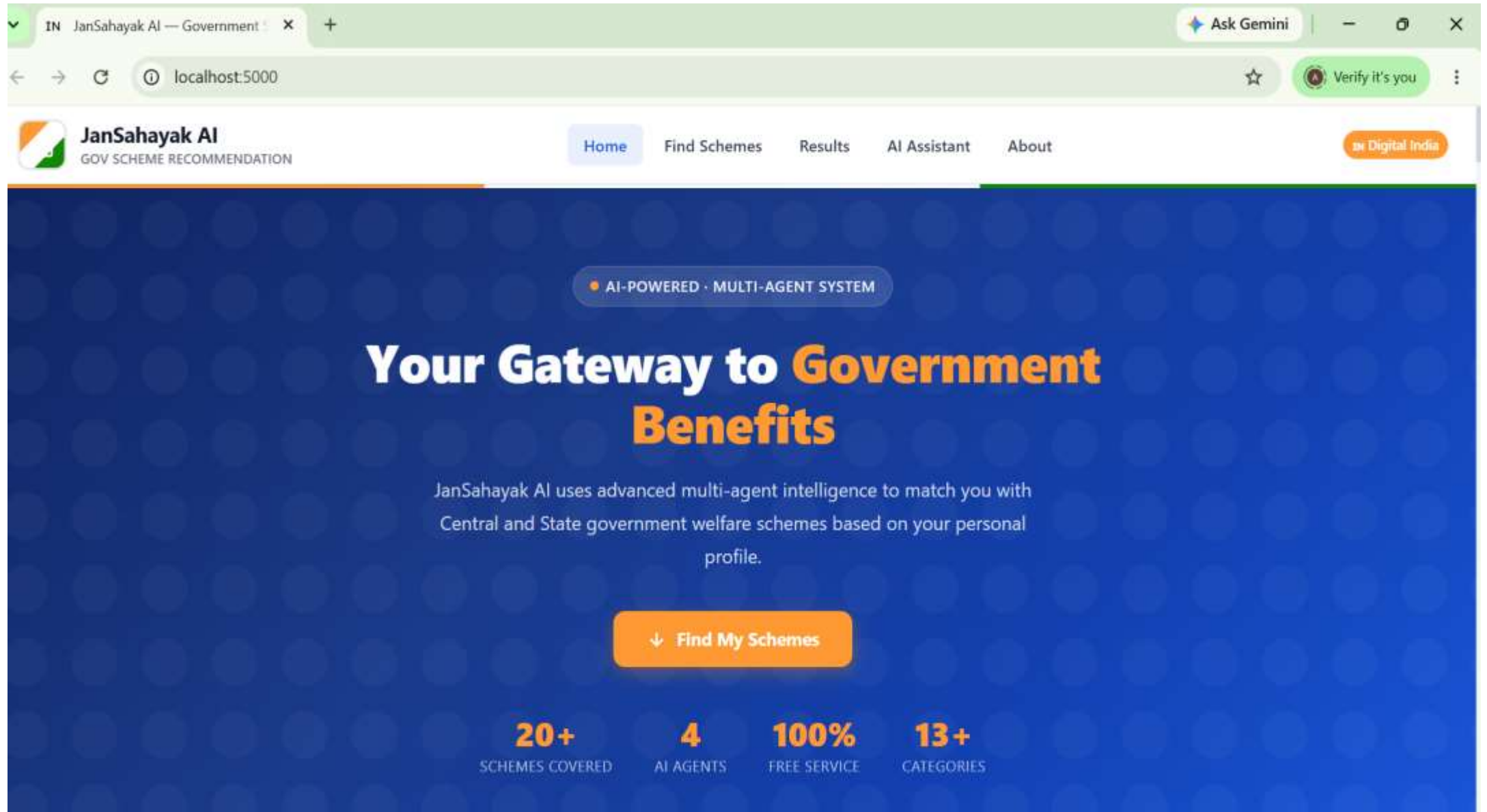
Before Prompt



After the Prompt



User Interface for JanSahayak AI



User Prompt for JanSahayak AI



JanSahayak AI
GOV SCHEME RECOMMENDATION

Home

Find Schemes

Results

AI Assistant

About

Get Digital India

STEP 1 OF 1

Your Citizen Profile

Fill in your details to get personalized government scheme recommendations.
All fields marked * are required.

 PERSONAL INFORMATION

Full Name *

rajesh

Age *

43

Gender *

Male

Marital Status

Married

 LOCATION DETAILS

JanSahayak AI Response



JanSahayak AI
GOV SCHEME RECOMMENDATION

Home

Find Schemes

Results

AI Assistant

About

Digital India

**Privacy First:** Your data is processed locally to generate recommendations only. We do not store, share, or transmit your personal information to any third party. This service is purely for guidance purposes.

PERSONALIZED RECOMMENDATIONS

Hello, rajesh! Here are your personalized scheme recommendations.

Economic Category: EWS

Edit Profile

17

Total Eligible Schemes

12

Schemes Shown

OBC, male, rural

Social Category

Free

Application Cost



 AI Recommendation Summary

Based on your profile, we have found 17 government schemes you may be eligible for. The top recommendation is ****Sukanya Samriddhi Yojana (SSY)**** which provides 8.2% interest rate, tax exemption under 80C, maturity on girl turning 21. Please review the schemes below and proceed with the application steps provided.

Novelty and Uniqueness

JanSahayak AI combines **Agentic AI, Multi-Agent Intelligence, RAG, and IBM watsonx.ai** to deliver a smart, personalized, and proactive Government Scheme Recommendation System.

- ❑ **Multi-Agent AI** – Specialized agents collaborate for citizen profiling, eligibility verification, scheme recommendation, document assistance, and application guidance.
- 🎯 **Personalized Recommendations** – Suggests schemes based on age, income, education, occupation, location, category, and citizen needs.
- ✓ **Intelligent Eligibility Verification** – Automatically identifies eligible schemes, eliminating manual searches across multiple government websites.
- 🌐 **Multilingual & Explainable AI** – Provides recommendations in regional languages with clear eligibility reasons, required documents, and application steps.
- ☁️ **Enterprise-Ready Platform** – Built on **IBM BOB, IBM watsonx.ai, IBM Granite Models, and IBM Cloud** for secure, scalable, and governed AI deployment.

JanSahayak AI acts as a **smart digital citizen assistant**, offering end-to-end personalized government scheme discovery, eligibility verification, and application support—not just a simple information portal.

Git Hub URL:

<https://github.com/Manjulasudhakar/JanSahayak-AI.git>

1. Generative AI in Action credly certificate
2. IBM BOB Certificate
3. Project PPT
4. Problem Statement

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Future Scope

➤ Integration with DigiLocker & Aadhaar

Integrate with **DigiLocker** and Aadhaar-based verification (where legally permitted) to automatically retrieve required documents, simplify eligibility verification, and accelerate the application process.

➤ AI Voice Assistant & Multilingual Support

Enhance the platform with a **voice-enabled AI assistant** supporting multiple Indian languages, enabling citizens—especially those in rural areas—to interact naturally and access government schemes more easily.

➤ Real-Time Government Scheme Updates

Connect with official government portals to automatically retrieve newly launched schemes, policy changes, eligibility updates, and application deadlines using **RAG**, ensuring citizens always receive the latest information.

➤ Mobile App & WhatsApp Integration

Develop Android/iOS applications and integrate with **WhatsApp** to deliver personalized scheme recommendations, application status updates, and reminder notifications directly to citizens.

Certificates Earned:

Generative AI in Action Credly Certificate



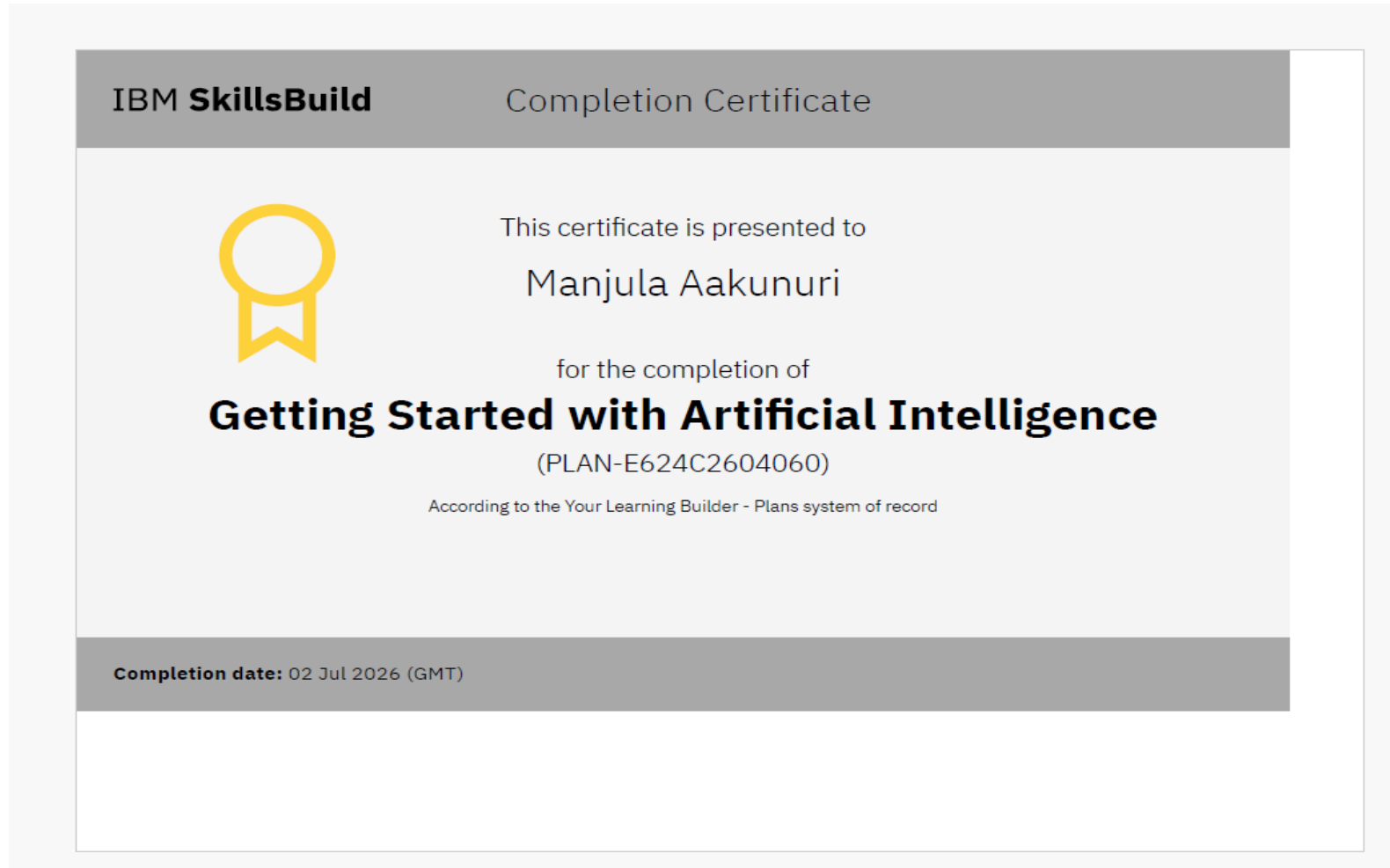
Certificates Earned:

IBM BOB Certificate



Certificates Earned:

Getting Started with Artificial Intelligence Certificate



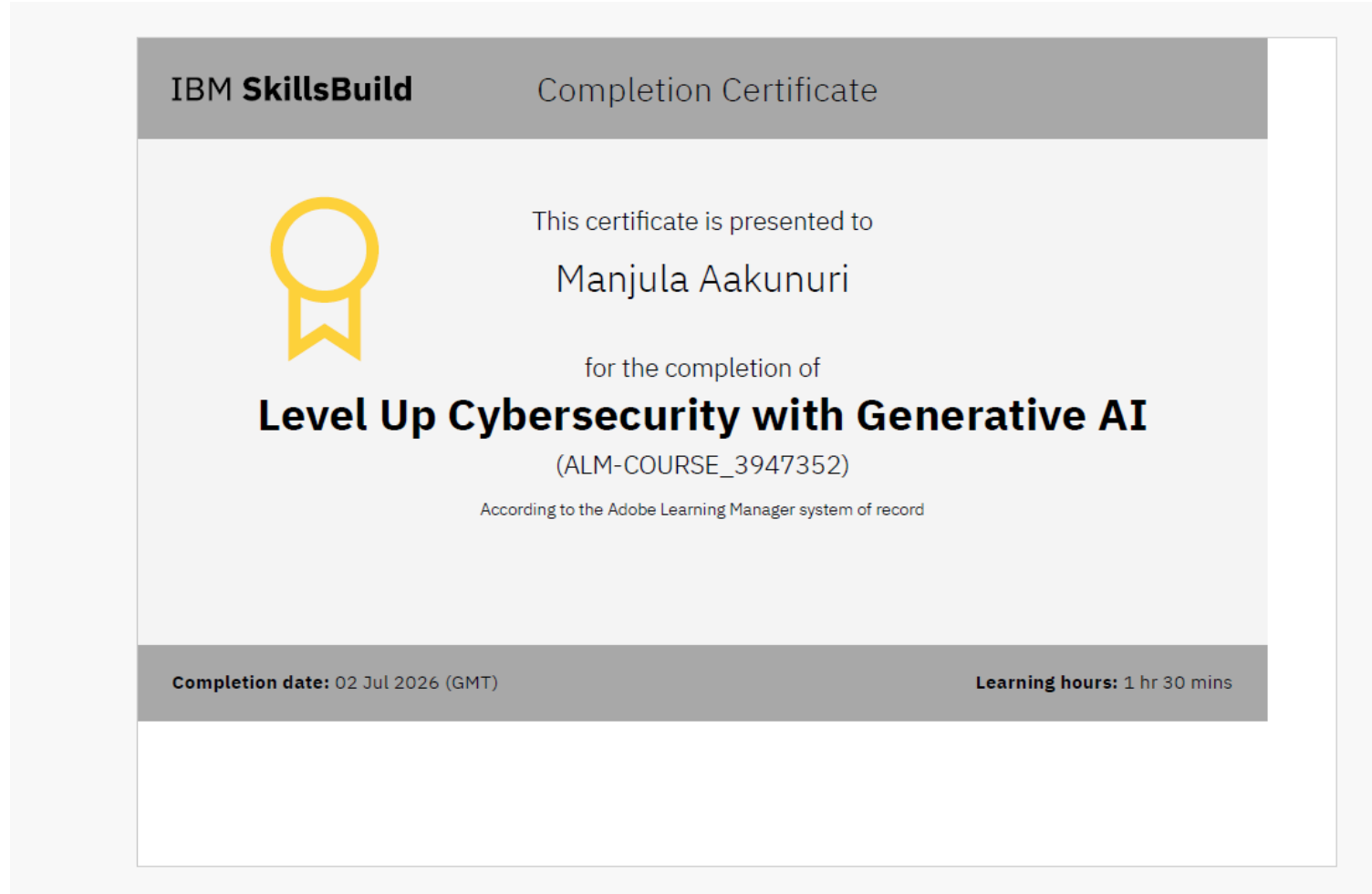
Certificates Earned:

Getting Started with Cyber Security Certificate



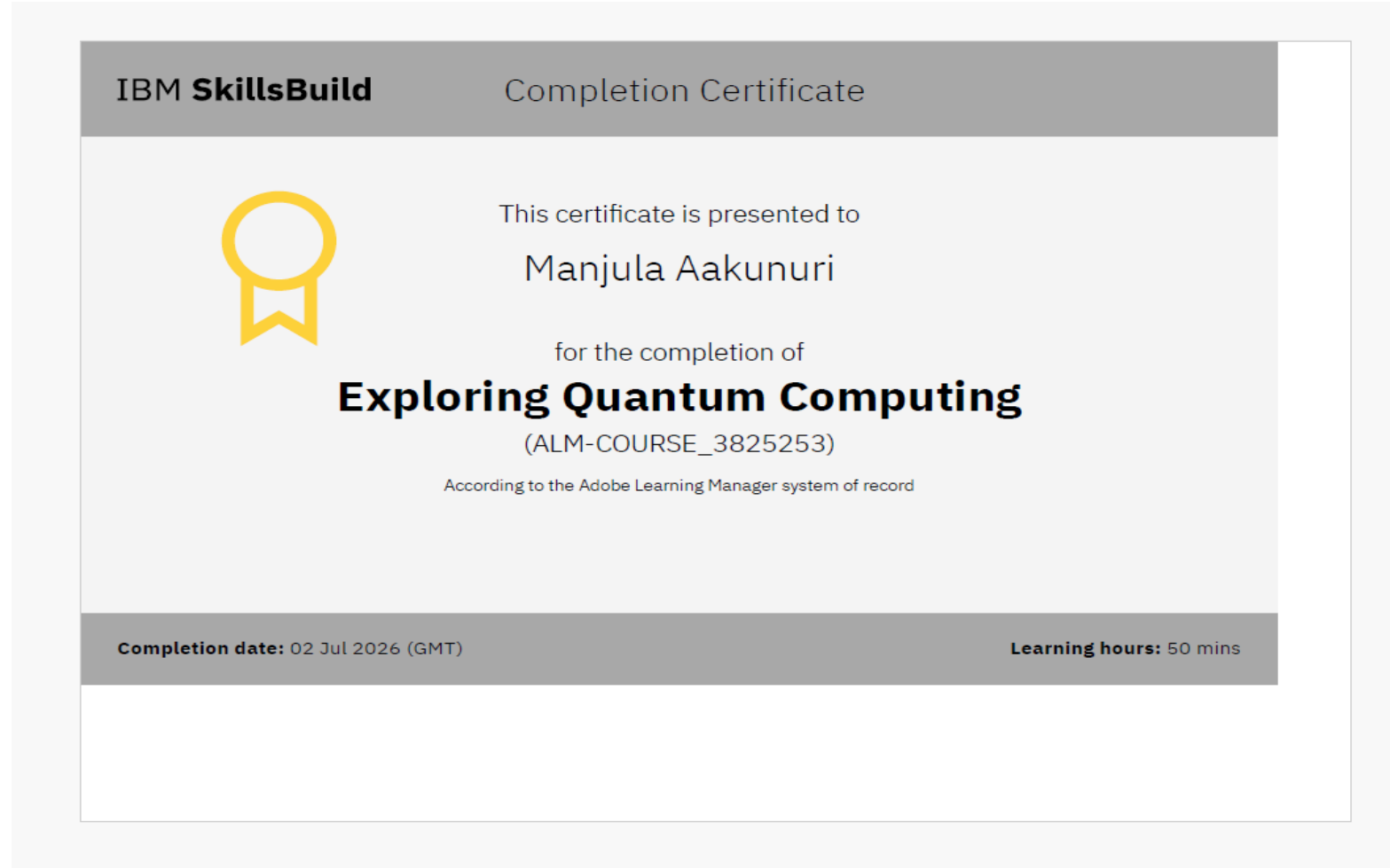
Certificates Earned:

Level Up Cyber Security with Generative AI Certificate



Certificates Earned:

Exploring Quantum Computing Certificate



IBM SKILLS BUILD PROFILE

Profile



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Thank You!

Thank you for your time and interest.